StudySync class diagram

Static structure of the MMDC Student Groupmate Matching System: 14 classes and 9 enumerations, with encapsulated attributes, constructors, accessors, and domain operations in standard UML notation.

GENERALIZATION · COMPOSITION · ASSOCIATION · NAVIGABLE END · ASSOCIATION CLASS · DEPENDENCY

\- private · + public · «create» constructor · {abstract} abstract member · 0..* multiplicity · italic role name

«use»

Administrator

-adminCode: String

+«create» +Administrator(userId: int, fullName: String, email: String, passwordHash: String, adminCode: String)
+getAdminCode(): String
+setAdminCode(adminCode: String): void
+getRole(): String
+manageUser(userId: int, action: String): void
+manageSubject(subjectId: int, action: String): void
+reviewReport(reportId: int, resolution: String): void

reviewer 0..1

User
{abstract}

-userId: int
-fullName: String
-email: String
-passwordHash: String
-accountStatus: AccountStatus = ACTIVE

+«create» +User(userId: int, fullName: String, email: String, passwordHash: String)
+getUserId(): int
+setUserId(userId: int): void
+getFullName(): String
+setFullName(fullName: String): void
+getEmail(): String
+setEmail(email: String): void
+getPasswordHash(): String
+setPasswordHash(passwordHash: String): void
+getAccountStatus(): AccountStatus
+setAccountStatus(accountStatus: AccountStatus): void
+login(email: String, password: String): boolean
+logout(): void
+updateBasicInfo(fullName: String, email: String): void
+getRole(): String {abstract}

1 reportedUser

Notification

-notificationId: int
-type: NotificationType
-message: String
-read: boolean = false
-createdAt: LocalDateTime

+«create» +Notification(notificationId: int, type: NotificationType, message: String, createdAt: LocalDateTime)
+getNotificationId(): int
+setNotificationId(notificationId: int): void
+getType(): NotificationType
+setType(type: NotificationType): void
+getMessage(): String
+setMessage(message: String): void
+isRead(): boolean
+setRead(read: boolean): void
+getCreatedAt(): LocalDateTime
+setCreatedAt(createdAt: LocalDateTime): void
+markAsRead(): void

0..* 1

MatchingService

-minimumScore: double
-maximumResults: int

+«create» +MatchingService(minimumScore: double, maximumResults: int)
+getMinimumScore(): double
+setMinimumScore(minimumScore: double): void
+getMaximumResults(): int
+setMaximumResults(maximumResults: int): void
+findCandidates(studentId: int, subjectId: int): Student[0..*]
+calculateCompatibility(studentId: int, candidateId: int): double
+generateRecommendations(studentId: int, subjectId: int): MatchRecommendation[0..*]

«create»

MatchRecommendation

-recommendationId: int
-compatibilityScore: double
-status: RecommendationStatus = NEW
-createdAt: LocalDateTime

+«create» +MatchRecommendation(recommendationId: int, compatibilityScore: double, createdAt: LocalDateTime)
+getRecommendationId(): int
+setRecommendationId(recommendationId: int): void
+getCompatibilityScore(): double
+setCompatibilityScore(compatibilityScore: double): void
+getStatus(): RecommendationStatus
+setStatus(status: RecommendationStatus): void
+getCreatedAt(): LocalDateTime
+setCreatedAt(createdAt: LocalDateTime): void
+markViewed(): void
+dismiss(): void

0..* recipient
0..* candidate
0..*

UserReport

-reportId: int
-reason: String
-details: String
-status: ReportStatus = PENDING
-submittedAt: LocalDateTime
-resolution: String

+«create» +UserReport(reportId: int, reason: String, details: String, submittedAt: LocalDateTime)
+getReportId(): int
+setReportId(reportId: int): void
+getReason(): String
+setReason(reason: String): void
+getDetails(): String
+setDetails(details: String): void
+getStatus(): ReportStatus
+setStatus(status: ReportStatus): void
+getSubmittedAt(): LocalDateTime
+setSubmittedAt(submittedAt: LocalDateTime): void
+getResolution(): String
+setResolution(resolution: String): void
+resolve(resolution: String): void

0..* reporter

Student

-studentNumber: String

+«create» +Student(userId: int, fullName: String, email: String, passwordHash: String, studentNumber: String)
+getStudentNumber(): String
+setStudentNumber(studentNumber: String): void
+getRole(): String
+viewPotentialGroupmates(subjectId: int): MatchRecommendation[0..*]
+sendGroupmateRequest(receiverId: int, subjectId: int): GroupmateRequest
+respondToRequest(requestId: int, accept: boolean): void
+createProjectGroup(subjectId: int, groupName: String): ProjectGroup
+submitReport(reportedUserId: int, reason: String): UserReport

1 sender
1 receiver
1..* member
0..*

GroupmateRequest

-requestId: int
-status: RequestStatus = PENDING
-message: String
-sentAt: LocalDateTime
-respondedAt: LocalDateTime

+«create» +GroupmateRequest(requestId: int, message: String, sentAt: LocalDateTime)
+getRequestId(): int
+setRequestId(requestId: int): void
+getStatus(): RequestStatus
+setStatus(status: RequestStatus): void
+getMessage(): String
+setMessage(message: String): void
+getSentAt(): LocalDateTime
+setSentAt(sentAt: LocalDateTime): void
+getRespondedAt(): LocalDateTime
+setRespondedAt(respondedAt: LocalDateTime): void
+accept(): void
+decline(): void
+cancel(): void

Subject

-subjectId: int
-subjectCode: String
-subjectName: String
-term: String
-active: boolean = true

+«create» +Subject(subjectId: int, subjectCode: String, subjectName: String, term: String)
+getSubjectId(): int
+setSubjectId(subjectId: int): void
+getSubjectCode(): String
+setSubjectCode(subjectCode: String): void
+getSubjectName(): String
+setSubjectName(subjectName: String): void
+getTerm(): String
+setTerm(term: String): void
+isActive(): boolean
+setActive(active: boolean): void
+updateDetails(subjectCode: String, subjectName: String, term: String): void

0..* 1
1 0..*

Availability

-availabilityId: int
-dayOfWeek: String
-startTime: LocalTime
-endTime: LocalTime
-recurring: boolean = true

+«create» +Availability(availabilityId: int, dayOfWeek: String, startTime: LocalTime, endTime: LocalTime, recurring: boolean)
+getAvailabilityId(): int
+setAvailabilityId(availabilityId: int): void
+getDayOfWeek(): String
+setDayOfWeek(dayOfWeek: String): void
+getStartTime(): LocalTime
+setStartTime(startTime: LocalTime): void
+getEndTime(): LocalTime
+setEndTime(endTime: LocalTime): void
+isRecurring(): boolean
+setRecurring(recurring: boolean): void
+overlaps(other: Availability): boolean
+getDurationInHours(): double

0..* 1

CollaborationProfile

-profileId: int
-itMajor: String
-employmentStatus: EmploymentStatus
-workSchedule: String
-workingStyle: WorkingStyle
-profileSummary: String
-lastUpdated: LocalDateTime

+«create» +CollaborationProfile(profileId: int, itMajor: String, employmentStatus: EmploymentStatus, workingStyle: WorkingStyle)
+getProfileId(): int
+setProfileId(profileId: int): void
+getItMajor(): String
+setItMajor(itMajor: String): void
+getEmploymentStatus(): EmploymentStatus
+setEmploymentStatus(employmentStatus: EmploymentStatus): void
+getWorkSchedule(): String
+setWorkSchedule(workSchedule: String): void
+getWorkingStyle(): WorkingStyle
+setWorkingStyle(workingStyle: WorkingStyle): void
+getProfileSummary(): String
+setProfileSummary(profileSummary: String): void
+getLastUpdated(): LocalDateTime
+setLastUpdated(lastUpdated: LocalDateTime): void
+updateProfile(itMajor: String, employmentStatus: EmploymentStatus, workSchedule: String, workingStyle: WorkingStyle): void
+addAvailability(availability: Availability): void
+removeAvailability(availabilityId: int): void

GroupMembership

-membershipId: int
-role: MembershipRole = MEMBER
-joinedAt: LocalDateTime
-active: boolean = true

+«create» +GroupMembership(membershipId: int, role: MembershipRole, joinedAt: LocalDateTime)
+getMembershipId(): int
+setMembershipId(membershipId: int): void
+getRole(): MembershipRole
+setRole(role: MembershipRole): void
+getJoinedAt(): LocalDateTime
+setJoinedAt(joinedAt: LocalDateTime): void
+isActive(): boolean
+setActive(active: boolean): void
+leaveGroup(): void

{members <= maximumMembers}

ProjectGroup

-groupId: int
-groupName: String
-maximumMembers: int
-createdAt: LocalDateTime
-status: GroupStatus = FORMING

+«create» +ProjectGroup(groupId: int, groupName: String, maximumMembers: int, createdAt: LocalDateTime)
+getGroupId(): int
+setGroupId(groupId: int): void
+getGroupName(): String
+setGroupName(groupName: String): void
+getMaximumMembers(): int
+setMaximumMembers(maximumMembers: int): void
+getCreatedAt(): LocalDateTime
+setCreatedAt(createdAt: LocalDateTime): void
+getStatus(): GroupStatus
+setStatus(status: GroupStatus): void
+addMember(studentId: int): void
+removeMember(studentId: int): void
+viewMembers(): GroupMembership[1..*]
+isFull(): boolean

0..* 0..*

SubjectEnrollment

-enrollmentId: int
-lookingForGroup: boolean = true
-enrolledAt: LocalDate

+«create» +SubjectEnrollment(enrollmentId: int, lookingForGroup: boolean, enrolledAt: LocalDate)
+getEnrollmentId(): int
+setEnrollmentId(enrollmentId: int): void
+isLookingForGroup(): boolean
+setLookingForGroup(lookingForGroup: boolean): void
+getEnrolledAt(): LocalDate
+setEnrolledAt(enrolledAt: LocalDate): void

use

use

«enumeration» AccountStatus

ACTIVE
SUSPENDED
DEACTIVATED

«enumeration» EmploymentStatus

FULL_TIME_STUDENT
WORKING_STUDENT

«enumeration» WorkingStyle

INDEPENDENT
FREQUENT_COLLABORATION
STRUCTURED_MEETINGS
FLEXIBLE

«enumeration» RecommendationStatus

NEW
VIEWED
DISMISSED

«enumeration» RequestStatus

PENDING
ACCEPTED
DECLINED
CANCELLED

«enumeration» GroupStatus

FORMING
COMPLETE
DISBANDED

«enumeration» MembershipRole

LEADER
MEMBER

«enumeration» NotificationType

MATCH_FOUND
REQUEST_RECEIVED
REQUEST_ACCEPTED
REQUEST_DECLINED
GROUP_INVITATION
GROUP_UPDATE

«enumeration» ReportStatus

PENDING
UNDER_REVIEW
RESOLVED
DISMISSED